MATHEMATICS (BCA-103)

ASSIGNMENT 3

1. Find $\frac{dy}{dx}$ when $y = \frac{x^2 + x + 1}{x + 5}$.

2. Find $\frac{dy}{dx}$ when $y = \frac{x^2 + x + 1}{\sqrt{x}}$.

3. Find $\frac{dy}{dx}$ when $y = a^{(3x-10)}$.

4. Find $\frac{dy}{dx}$ when $y = \frac{\sqrt{x-1}}{\sqrt{x+1}}$.

5. Find $\frac{dy}{dx}$ when $y = \cot^{-1}(\operatorname{cosec} x + \cot x)$.

6. Find $\frac{dy}{dx}$ when $y = \tan^{-1}\left(\frac{\sqrt{1+x^2}+1}{x}\right)$.

7. Find $\frac{dy}{dx}$ when $y = x^x + (\log x)^x$.

8. Find $\frac{dy}{dx}$ when $y = \cot 3x$.

9. Find $\frac{dy}{dx}$ when $y = \sin^2 x^3$.

10. Find $\frac{dy}{dx}$ when $y = \cot^{-1} x^3$.

11. Differentiate $\sqrt{1+x^2}$ w.r.t. x.

12. Differentiate $\cos^2 x$ w.r.t. x.

13. Differentiate $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ w.r.t. $\tan^{-1} x$.

14. Differentiate $y = \frac{1-\sqrt{x}}{1+\sqrt{x}}$ w.r.t. x.

15. Differentiate $\frac{\sin x - x^2}{\cot 2x}$ w.r.t. x.

16. Differentiate $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ w.r.t. x.

17. Differentiate $(\sin^{-1} x)^x$ w.r.t. x.

18. If $y = x + \sqrt{x^2 - 1}$, prove that $(y - x)\frac{dy}{dx} - y = 0$.

19. Find the differential coefficient of $\tan x$ by first principle.

20. Differentiate w.r.t. x (i) $\frac{x}{\sin 3x}$ (ii) $\frac{x^2 + 1}{x + 1}$

21. Differentiate w.r.t. x (i) $\sqrt{\frac{1 - \sin x}{1 + \sin x}}$ (ii) $\tan^{-1}\left(\frac{\sqrt{1 + x^2} - 1}{x}\right)$

(iii) $x^{\log x}$ (iv) $\left(\frac{x\sqrt{1 + x^2}}{(x + 1)^{2/3}}\right)$

22. Differentiate w.r.t. x (i) $(2x + 3)\sqrt{x}$ (ii) $\frac{x^2 - 1}{x^2 + 7x + 1}$

(iii) $\frac{\sin x + x^2}{\cot 2x}$ (iv) $\sin^{-1}\left(x^{3/2}\right)$

23. Find $\frac{dy}{dx}$ where (i) $y = \sin^{-1}\sqrt{\frac{1 + x^2}{2}}$ (ii) $y = \log\sqrt{\frac{1 - \cos x}{1 + \cos x}}$

(iii) $y = x^x$ (iv) $x = \frac{1 - t^2}{1 + t^2}$, $y = \frac{2t}{1 + t^2}$

24. Differentiate w.r.t. x (i) $\sqrt{\frac{\sec x - 1}{\sec x + 1}}$ (ii) $\tan^{-1}\left(\frac{\sqrt{1 + x^2} + 1}{x}\right)$

(iii) $\left(x + \frac{1}{x}\right)\left(x^2 - \frac{1}{x^2}\right)$ (iv) $(2x + 3)^5 (3 - x)^4$